

Representation of  
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## Matrices

### Linear Algebra

Morales Carmona Ronny

Autumn 2026

## Definition

Let us choose a basis  $B_{\mathcal{V}} = \{|a_i\rangle\}_{i=1}^N$  of a linear space  $\mathcal{V}_N$ , and express an arbitrary vector  $|x\rangle$  in this basis:  $|x\rangle = \xi_j |a_j\rangle$ . We write

$$x = \begin{pmatrix} \xi_1 \\ \xi_2 \\ \vdots \\ \xi_N \end{pmatrix}$$

and say that the column vector  $x$  **represents**  $|x\rangle$  in  $B_{\mathcal{V}}$ .

## Remark

We can also have a linear homomorphism  $\mathbf{A} \in \text{Hom}(\mathcal{V}_N, \mathcal{W}_M)$  act on the basis vectors in  $B_{\mathcal{V}}$  to give vectors in the  $M$ -dimensional linear space  $\mathcal{W}_M$ :  $|w_k\rangle = \mathbf{A}|a_k\rangle$ . The latter can be written as a linear combination of basis vectors  $B_{\mathcal{W}} = \{|b_j\rangle\}_{j=1}^M$  in  $\mathcal{W}_M$ :

$$|w_\ell\rangle = \alpha_{j\ell}|b_j\rangle.$$

The components can be arranged in a column as before to give a representation of the corresponding vectors:

$$\mathbf{w}_1 = \begin{pmatrix} \alpha_{11} \\ \alpha_{21} \\ \vdots \\ \alpha_{M1} \end{pmatrix}, \quad \mathbf{w}_2 = \begin{pmatrix} \alpha_{12} \\ \alpha_{22} \\ \vdots \\ \alpha_{M2} \end{pmatrix}, \quad \dots, \quad \mathbf{w}_N = \begin{pmatrix} \alpha_{1N} \\ \alpha_{2N} \\ \vdots \\ \alpha_{MN} \end{pmatrix}.$$

The operator itself is determined by the collection of all these vectors, i.e., by a matrix.

## Definition

Let  $\mathbf{A} \in \text{Hom}(\mathcal{V}_N, \mathcal{W}_M)$  and let  $B_{\mathcal{V}} = \{|a_i\rangle\}_{i=1}^N, B_{\mathcal{W}} = \{|b_j\rangle\}_{j=1}^M$  be bases for  $\mathcal{V}$  and  $\mathcal{W}$ , respectively. Define  $\{\alpha_{ji}\}_{j=1,i=1}^{M,N}$  by  $\mathbf{A}|a_i\rangle = \alpha_{ji}|b_j\rangle$ . Then we write

$$\mathbf{A} = \begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1N} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2N} \\ \vdots & \vdots & & \vdots \\ \alpha_{M1} & \alpha_{M2} & \cdots & \alpha_{MN} \end{pmatrix}$$

and call  $\mathbf{A}$  the **matrix representing  $\mathbf{A}$  in bases  $B_{\mathcal{V}}$  and  $B_{\mathcal{W}}$** .

## Example 4.1

Find the matrix representation of the operator  $\mathbf{T} \in \text{End}(\mathbb{R}^2)$  defined by

$$\mathbf{T} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + y \\ x - y \end{pmatrix}$$

in the canonical basis and in the basis  $B$  given by

$$B = \left\{ \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}.$$

## Theorem 4.2

*The two linear spaces  $\text{Hom}(\mathcal{V}_N, \mathcal{W}_M)$  and  $\mathcal{M}^{M \times N}$  are isomorphic. An explicit isomorphism is established only when a basis is chosen for each linear space, in which case, an operator is identified with its matrix representation.*

## Notation

Let  $\mathbf{A} \in \text{Hom}(\mathcal{V}_N, \mathcal{W}_M)$ . Choose bases  $B_{\mathcal{V}}$  for  $\mathcal{V}$  and  $B_{\mathcal{W}}$ . We denote the matrix representing  $\mathbf{A}$  in these bases by  $\mathbf{M}_{B_{\mathcal{V}}}^{B_{\mathcal{W}}}(\mathbf{A})$ , where

$$\mathbf{M}_{B_{\mathcal{V}}}^{B_{\mathcal{W}}} : \text{Hom}(\mathcal{V}_N, \mathcal{W}_M) \rightarrow \mathcal{M}^{M \times N}$$

is the basis-dependent linear isomorphism. When  $\mathcal{V} = \mathcal{W}$ , we leave out the subscripts and superscripts of  $\mathbf{M}$ , keeping in mind that all matrices are representations in a single basis.

## Remark

Given the linear homomorphisms  $\mathbf{A} : \mathcal{V}_N \rightarrow \mathcal{W}_M$  and  $\mathbf{B} : \mathcal{W}_M \rightarrow \mathcal{U}_K$ , we can form the composite  $\mathbf{B} \circ \mathbf{A} : \mathcal{V}_N \rightarrow \mathcal{U}_K$ . We can also choose bases  $B_{\mathcal{V}} = \{|a_i\rangle\}_{i=1}^N$ ,  $B_{\mathcal{W}} = \{|b_i\rangle\}_{i=1}^M$ ,  $B_{\mathcal{U}} = \{|c_i\rangle\}_{i=1}^K$  for  $\mathcal{V}$ ,  $\mathcal{W}$ , and  $\mathcal{U}$ , respectively. Then  $\mathbf{A}$ ,  $\mathbf{B}$ , and  $\mathbf{B} \circ \mathbf{A}$  will be represented by an  $M \times N$ , a  $K \times M$ , and a  $K \times N$  matrix, respectively, and we have

$$\mathbf{M}_{B_{\mathcal{V}}}^{B_{\mathcal{U}}}(\mathbf{B} \circ \mathbf{A}) = \mathbf{M}_{B_{\mathcal{W}}}^{B_{\mathcal{U}}}(\mathbf{B})\mathbf{M}_{B_{\mathcal{V}}}^{B_{\mathcal{W}}}(\mathbf{A}),$$

where on the right-hand side the product is defined as the usual product of matrices. If  $\mathcal{V} = \mathcal{W} = \mathcal{U}$ , we write the previous equation as

$$\mathbf{M}(\mathbf{B} \circ \mathbf{A}) = \mathbf{M}(\mathbf{B})\mathbf{M}(\mathbf{A}).$$

## Notation

*Matrices are determined entirely by their elements. For this reason a matrix  $A$  whose elements are  $\alpha_{11}, \alpha_{12}, \dots$  is sometimes denoted by  $(\alpha_{ij})$ . Similarly, the elements of this matrix are denoted by  $(A)_{ij}$ . In the context of this notation, therefore, we can write*

$$(A + B)_{ij} = (A)_{ij} + (B)_{ij} \implies (\alpha_{ij} + \beta_{ij}) = (\alpha_{ij}) + (\beta_{ij}),$$

$$(\gamma A)_{ij} = \gamma (A)_{ij} \implies \gamma (\alpha_{ij}) = (\gamma \alpha_{ij}),$$

$$(0)_{ij} = 0,$$

$$(1)_{ij} = \delta_{ij}.$$

## Definition

The **transpose** of an  $M \times N$  matrix  $A$  is an  $N \times M$  matrix  $A^t$  obtained by interchanging the rows and columns of  $A$ :

$$(A^t)_{ij} = (A)_{ji}, \quad \text{or} \quad (\alpha_{ij})^t = (\alpha_{ji}).$$

### Theorem 4.3

*Let  $A$  and  $B$  be two matrices for which the operation of addition and/or multiplication are defined. Then*

- i)  $(A + B)^t = A^t + B^t;$
- ii)  $(AB)^t = B^t A^t;$
- iii)  $(A^t)^t = A.$

## Theorem 4.4

Let  $\mathbf{T} \in \text{End}(\mathcal{V}, \mathcal{W})$  and  $B_{\mathcal{V}}$  and  $B_{\mathcal{W}}$  be bases in  $\mathcal{V}$  and  $\mathcal{W}$ . Let  $\mathbf{T}^*$ ,  $B_{\mathcal{V}}^*$ , and  $B_{\mathcal{W}}^*$  be duals to  $\mathbf{T}$ ,  $B_{\mathcal{V}}$ , and  $B_{\mathcal{W}}$ , respectively. Let  $\mathbf{T} \doteq \mathbf{M}_{B_{\mathcal{V}}}^{B_{\mathcal{W}}}(\mathbf{T})$  and  $\mathbf{T}^* \doteq \mathbf{M}_{B_{\mathcal{W}}^*}^{B_{\mathcal{V}}^*}(\mathbf{T}^*)$ . Then  $\mathbf{T}^* = \mathbf{T}^t$ .

## Definition

A matrix  $S$  is **symmetric** if  $S^t = S$ . Similarly, a matrix  $A$  is **antisymmetric** if  $A^t = -A$ .

## Remark

*Any matrix  $A$  can be written as*

$$A = \frac{1}{2} (A + A^t) + \frac{1}{2} (A - A^t),$$

*where the first term is symmetric and the second is antisymmetric.*

## Definition

*A (real) matrix satisfying  $A^t A = A A^t = 1$  is called **orthogonal**.*

## Definition

**Complex conjugation** is an operations under which all elements of a matrix are complex conjugated. Denoting the complex conjugate of  $A$  by  $A^*$ , we have  $(A^*)_{ij} = (A)_{ij}^*$ , or  $(\alpha_{ij})^* = (\alpha_{ij}^*)$ .

## Definition

The **hermitian conjugate** of a matrix  $A$  is the complex conjugation of the transpose, or, equivalently, the transpose of the complex conjugation. Thus, we have

$$A^\dagger = (A^t)^* = (A^*)^t,$$
$$(A^\dagger)_{ij} = (A)_{ji}^* \quad \text{or} \quad (\alpha_{ij})^\dagger = (\alpha_{ji}^*).$$

## Definition

A **hermitian** matrix  $H$  satisfies  $H^\dagger = H$ , or, in terms of elements,

$$\eta_{ji}^* = \eta_{ij}.$$

A **unitary** matrix satisfies  $UU^\dagger = U^\dagger U = 1$ , or, in terms of elements,

$$\mu_{ik}\mu_{jk}^* = \mu_{ki}^*\mu_{kj} = \delta_{ij}.$$

## Remark

- i) *The diagonal elements of a hermitian matrix are real.*
- ii) *The  $k$ th column of a hermitian matrix is the complex conjugate of its  $k$ th row, and vice versa.*
- iii) *A real hermitian matrix is symmetric.*
- iv) *The rows of an  $N \times N$  unitary matrix, when considered as vectors in  $\mathbb{C}^N$ , form an orthonormal set under the usual inner product, as do the columns.*
- v) *A real unitary matrix is orthogonal.*

## Definition

A matrix in which all of its off-diagonal elements are zero is called a **diagonal matrix**. A diagonal matrix whose diagonal elements are  $\{\lambda_k\}_{k=1}^N$  is denoted by  $\text{diag}(\lambda_1, \lambda_2, \dots, \lambda_N) \equiv \text{diag}(\lambda_k)_{k=1}^N$ .

## Example 4.5

Let  $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$  be a diagonal matrix, and  $f(x)$  a function that has a Taylor series expansion around zero. Show that

$$f(D) = \text{diag}(f(\lambda_1), f(\lambda_2), \dots, f(\lambda_n)).$$

We list some familiar matrices in physics.

- i) A prototypical symmetric matrix is that of the moment of inertia encountered in mechanics. The  $ij$ th element of this matrix is defined as  $I_{ij} = \int d^3x \rho(x_1, x_2, x_3) x_i x_j$ , where  $x_i$  is the  $i$ th Cartesian coordinate of a point in the distribution of mass described by the volume density  $\rho(x_1, x_2, x_3)$ . It is clear that  $I_{ij} = I_{ji}$ , or  $I = I^t$ . The moment of inertia matrix can be represented as

$$I = \begin{pmatrix} I_{11} & I_{12} & I_{13} \\ I_{12} & I_{22} & I_{23} \\ I_{13} & I_{23} & I_{33} \end{pmatrix}$$

- ii) An example of an antisymmetric matrix is the electromagnetic field tensor given by

$$F = \begin{pmatrix} 0 & -B_3 & B_2 & E_1 \\ B_3 & 0 & -B_1 & E_2 \\ -B_2 & B_1 & 0 & E_3 \\ -E_1 & -E_2 & -E_3 & 0 \end{pmatrix}.$$

iii) Examples of hermitian matrices are the  $2 \times 2$  **Pauli spin matrices**:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

iv) The most frequently encountered orthogonal matrices are rotations. One of such matrix represents the rotation of a 3-dimensional rigid body in terms of **Euler angles** and is used in mechanics. Attaching a coordinate system to the body, a general rotation can be decomposed into a rotation of angle  $\varphi$  about the  $z$ -axis, followed by a rotation of angle  $\theta$  about the *new*  $x$ -axis, followed by a rotation of angle  $\psi$  about the *new*  $z$ -axis. We simply exhibit this matrix in terms of these angles:

$$\begin{pmatrix} \cos \psi \cos \varphi - \sin \psi \cos \theta \sin \varphi & -\cos \psi \sin \varphi - \sin \psi \cos \theta \cos \varphi & \sin \psi \sin \theta \\ \sin \psi \cos \varphi + \cos \psi \cos \theta \sin \varphi & -\sin \psi \sin \varphi + \cos \psi \cos \theta \cos \varphi & -\cos \psi \sin \theta \\ \sin \theta \sin \varphi & \sin \theta \cos \varphi & \cos \theta \end{pmatrix}.$$

## Remark

The matrix representation of  $\mathbf{A} \in \text{End}(\mathcal{V})$  is facilitated by choosing an orthonormal basis  $B = \{|e_i\rangle\}_{i=1}^N$ . The matrix elements of  $\mathbf{A}$  can be found in such a basis by  $(A)_{ij} = \alpha_{ij} = \langle e_i | \mathbf{A} | e_j \rangle$ . We can also show that in an orthonormal basis, the  $i$ th component  $\xi_i$  of a vector is found by multiplying the vector by  $\langle e_i |$ . This expression for  $\xi_i$  allows us to write the expansion of  $|x\rangle$  as

$$|x\rangle = \sum_{j=1}^N \underbrace{\langle e_j | x \rangle}_{\xi_j} |e_j\rangle = \sum_{j=1}^N |e_j\rangle \langle e_j | x \rangle \implies \mathbf{1} = \sum_{j=1}^N |e_j\rangle \langle e_j|,$$

which is the same as Theorem 3.27.

## Remark

We may now calculate the matrix representing the hermitian conjugate of an operator  $\mathbf{T}$ . In an orthonormal basis, the elements of this matrix are given by  $\tau_{ij} = \langle e_i | \mathbf{T} | e_j \rangle$ . Taking the complex conjugate of this equation and using the definition of  $\mathbf{T}^\dagger$  we get

$$\tau_{ij} = \langle e_i | \mathbf{T} | e_j \rangle^\dagger = \langle e_j | \mathbf{T} | e_i \rangle, \quad \text{or} \quad (\mathbf{T}^\dagger) = \tau_{ji}^*.$$

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## Remark

*Only in an orthonormal basis is the adjoint of an operator represented by the adjoint of the matrix representing that operator. In particular, a hermitian operator is represented by a hermitian matrix only if an orthonormal basis is used.*

## Example 4.6

Consider the matrix representation of the hermitian operator  $\mathbf{H}$  in a general—not orthonormal—basis  $B = \{|a_i\rangle\}_{i=1}^N$ . The elements of the matrix corresponding to  $\mathbf{H}$  are given by

$$\mathbf{H}|a_k\rangle = \sum_{j=1}^N \eta_{jk} |a_j\rangle, \quad \text{or} \quad \mathbf{H}|a_i\rangle = \sum_{j=1}^N \eta_{ji} |a_j\rangle.$$

By considering  $\langle a_i | \mathbf{H} | a_k \rangle^*$  deduce that  $\langle a_k | \mathbf{H} | a_i \rangle = \eta_{jk}^* \langle a_j | a_i \rangle$  and then, by multiplying the second equation by  $\langle a_k |$  deduce that  $\langle a_k | \mathbf{H} | a_i \rangle = \eta_{ji} \langle a_k | a_j \rangle$ . Conclude that  $\eta_{jk}^* \langle a_j | a_i \rangle = \eta_{ji} \langle a_k | a_j \rangle$ . Show that if the  $|a_i\rangle$ 's are orthonormal, then we can conclude that  $\eta_{ik}^* = \eta_{ki}$ , as expected of a hermitian matrix.

### Example 4.7

Let  $\mathbf{U}$  be a unitary operator, i.e.,  $\mathbf{U}\mathbf{U}^\dagger = \mathbf{1}$  and let  $\{|e_i\rangle\}$  be an orthonormal basis. In this case we have

$$\langle e_i | \mathbf{U}\mathbf{U}^\dagger | e_j \rangle = \langle e_i | \mathbf{1} | e_j \rangle = \delta_{ij}.$$

Insert the completeness relation  $\mathbf{1} = |e_k\rangle\langle e_k|$  between  $\mathbf{U}$  and  $\mathbf{U}^\dagger$  to deduce that  $\mu_{ik}\mu_{jk}^* = \delta_{ij}$ . Similarly, doing the same calculation for  $\mathbf{U}^\dagger\mathbf{U}$ , deduce that  $\mu_{ki}^*\mu_{kj} = \delta_{ij}$ .

## Remark

Given a basis  $B = \{|b_i\rangle\}_{i=1}^N$ , we can write an arbitrary vector  $|a\rangle$  with components  $\{\alpha_i\}_{i=1}^N$  in  $B$  as  $|a\rangle = \alpha_i |b_i\rangle$ . Now suppose that we change the basis to  $B' = \{|b'_j\rangle\}_{j=1}^N$ . We ask ourselves, how are the components of  $|a\rangle$  in  $B'$  related to those in  $B$ ?

## Definition

Let  $B, B'$  be bases for  $\mathcal{V}_N$ . Define the numbers  $\{\rho_{ji}\}$  via  $|b_i\rangle = \rho_{ji}|b_j\rangle$ . The matrix

$$R = \begin{pmatrix} \rho_{11} & \rho_{12} & \cdots & \rho_{1N} \\ \rho_{21} & \rho_{22} & \cdots & \rho_{2N} \\ \vdots & \vdots & & \vdots \\ \rho_{N1} & \rho_{N2} & \cdots & \rho_{NN} \end{pmatrix}$$

is called the **basis transformation matrix**.

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## Corollary 4.8

*The basis transformation matrix of any two bases  $B, B'$  is invertible.*

## Definition

Consider the equation  $|b\rangle = \mathbf{A}|a\rangle$ . This equation has a corresponding matrix equation  $\mathbf{b} = \mathbf{A}\mathbf{a}$ . The matrix  $\mathbf{A}'$  defined by  $\mathbf{b}' = \mathbf{A}'\mathbf{a}'$  satisfies following relations

$$\mathbf{R}^{-1}\mathbf{A}'\mathbf{R} = \mathbf{A}, \quad \text{or} \quad \mathbf{A}' = \mathbf{R}\mathbf{A}\mathbf{R}^{-1}.$$

This is called a **similarity transformation** on  $\mathbf{A}$ , and  $\mathbf{A}'$  is said to be **similar** to  $\mathbf{A}$ .

**Remark**

*Let  $B = \{|e_i\rangle\}_{i=1}^N$  and  $B' = \{|e'_i\rangle\}_{i=1}^N$  be two orthonormal bases of  $\mathcal{V}_N$ . Then the components of the transformation matrix  $R$  are given by the relation*

$$\langle e'_j | e_i \rangle = \rho_{ji}.$$

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## Theorem 4.9

*The matrix that transforms one orthonormal basis into another is necessarily unitary.*

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## Example 4.10

*Show that the similarity transform of a function of a matrix is the same function of the similarity transform of the matrix.*

## Definition

*If  $A$  is representing  $\mathbf{A}$  in some basis, then we set  $\det \mathbf{A} = \det A$  (recall that the definition of determinant was independent of the basis).*

## Remark

Let  $\mathbf{A}$  be a linear operator on  $\mathcal{V}$  and let  $\{|e_i\rangle\}$  be a basis of  $\mathcal{V}$  in which  $\mathbf{A}$  is represented by  $A$ . Then

$$\begin{aligned}
 \Delta_{\mathbf{A}}(|e_1\rangle, \dots, |e_N\rangle) &= \Delta(\mathbf{A}|e_1\rangle, \dots, |e_N\rangle) \\
 &= \Delta(\alpha_{i_1 1}|e_{i_1}\rangle, \dots, \alpha_{i_N N}|e_{i_N}\rangle) \\
 &= \alpha_{i_1 1} \dots \alpha_{i_N N} \Delta(|e_{i_1}\rangle, \dots, |e_{i_N}\rangle) \\
 &= \sum_{\pi \in S_N} \alpha_{\pi(1)1} \dots \alpha_{\pi(N)N} \Delta(|e_{\pi(1)}\rangle, \dots, |e_{\pi(N)}\rangle) \\
 &= \sum_{\pi \in S_N} \alpha_{\pi(1)1} \dots \alpha_{\pi(N)N} \epsilon_{\pi} \Delta(|e_1\rangle, \dots, |e_N\rangle)
 \end{aligned}$$

which by definition of determinant yields

$$\det \mathbf{A} = \det A = \sum_{\pi \in S_N} \alpha_{\pi(1)1} \dots \alpha_{\pi(N)N} \epsilon_{\pi} \equiv \sum_{\pi \in S_N} \epsilon_{\pi} \prod_{k=1}^N (A)_{\pi(k)k}.$$

Noting that  $\sum_{\pi} = \sum_{\pi^{-1}}$  and  $\epsilon_{\pi} = \epsilon_{\pi^{-1}}$  we have the following theorem.

**Theorem 4.11**

Let  $\mathcal{V}$  be an  $N$ -dimensional linear space and let  $\mathbf{A} \in \text{End}(\mathcal{V})$  and  $A$  its representation in any basis of  $\mathcal{V}$ . Then

$$\det \mathbf{A} = \det A = \sum_{\pi \in S_N} \epsilon_{\pi} \prod_{j=1}^N (A)_{\pi(j)j} = \epsilon_{i_1 i_2 \dots i_N} (A)_{i_1 1} \dots (A)_{i_N N}$$

$$\det \mathbf{A} = \det A^t = \sum_{\pi \in S_N} \epsilon_{\pi} \prod_{j=1}^N (A)_{j\pi(j)} = \epsilon_{i_1 i_2 \dots i_N} (A)_{1i_1} \dots (A)_{Ni_N}$$

where  $\epsilon_{i_1 i_2 \dots i_N} = \epsilon_{\sigma(1)\sigma(2)\dots\sigma(p)} = \epsilon_{\sigma}$  with  $i_k \equiv \sigma(k)$ .

## Theorem 4.12

*Let  $A$  be a square matrix. Then*

- i)  *$\det A$  is linear with respect to any row or column vector of  $A$ .*
- ii) *If any two rows or two columns of  $A$  are interchanged,  $\det A$  changes sign.*
- iii) *Adding a multiple of one row [respectively, column] of  $A$  to another row [respectively, column] of  $A$  does not change  $\det A$ .*
- iv)  *$\det A = 0$  if and only if the rows [respectively, columns] are linearly dependent.*

## Definition

Given any  $N \times N$  matrix  $A$ , the determinant of the  $(N - 1) \times (N - 1)$  matrix obtained from  $A$  by eliminating its  $i$ th row and  $j$ th column is called a **minor of order  $N - 1$**  and denoted by  $M_{ij}$ . The product  $(-1)^{i+j} M_{ij}$  is called the **cofactor** of  $(A)_{ij}$ , and denoted by  $(\text{cof } A)_{ij}$ .

### Theorem 4.13 (Laplace Expansion for the Determinant of a Matrix)

*Given any  $N \times N$  matrix  $A$ , the determinant  $\det A$  of  $A$  is given by the following expression:*

$$\det A = \sum_{j=1}^N (A)_{ji} (\text{cof } A)_{ji} = \sum_{j=1}^N (A)_{ij} (\text{cof } A)_{ij} \quad (\text{no sum on } i).$$

**Example 4.14**

*Use the Laplace expansion to find the determinant of the matrix*

$$A = \begin{pmatrix} 0 & i & -1 \\ 0 & 2 & -i \\ 1+i & a & b \end{pmatrix}$$

*with  $a, b \in \mathbb{C}$ .*

**Theorem 4.15**

*The matrix  $A$  has an inverse if and only if  $\det A \neq 0$ . In this case we have*

$$A^{-1} = \frac{(\text{cof } A)^t}{\det A}.$$

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## Example 4.16

*Compute the inverse of a general  $2 \times 2$  matrix.*

## Definition

An **elementary row operation** on a matrix is one of the following:

- i) interchange of two rows of the matrix ,
- ii) multiplication of a row by a nonzero number, and
- iii) addition of a multiple of one row to another.

An **elementary column operation** is defined analogously.

## Definition

A matrix is in **triangular** (or **row-echelon**) form if it satisfies the following three conditions:

- i) Any row consisting of only zeros is below any row that contains at least one nonzero element.
- ii) Going from left to right, the first nonzero entry of any row is to the left of the first nonzero entry of any lower row.
- iii) The first nonzero entry of each row is 1.

**Theorem 4.17**

*For any invertible  $n \times n$  matrix  $A$ , the  $n \times 2n$  matrix  $(A|1)$  can be transformed into the  $n \times 2n$  matrix  $(1|A^{-1})^\dagger$  by means of a finite number of elementary row operations.*

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<sup>†</sup>The matrix  $(A|1)$  denotes the  $n \times 2n$  matrix obtained by juxtaposing the  $n \times n$  unit matrix to the right of  $A$ . It is easily shown that if  $A, B, C$  are  $n \times n$  matrices, then  $A(B|C) = (AB|AC)$ .

**Example 4.18**

*Evaluate the inverse of*

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & -2 \\ 2 & 1 & -1 \end{pmatrix}.$$

*Verify your result.*

## Definition

*Given any  $M \times N$  matrix  $A$ , an operator  $\mathbf{T}_A \in \text{Hom}(\mathcal{V}_N, \mathcal{W}_M)$  can be associated with  $A$ , and one can construct the kernel and the range of  $\mathbf{T}_A$ . The rank of  $\mathbf{T}_A$  is called the **rank** of  $A$ .*

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## Theorem 4.19

*The rank of a matrix is the dimension of the largest (square) submatrix whose determinant is not zero.*

## Definition

Let  $A$  be an  $N \times N$  matrix. The mapping  $\text{tr} : \mathcal{M}^{N \times N} \rightarrow \mathbb{C}$  given by  $\text{tr } A = \alpha_{ij} \delta_{ij}$  is called the **trace** of  $A$ .

**Theorem 4.20**

*The trace is a linear mapping. Furthermore,*

$$\operatorname{tr}(AB) = \operatorname{tr}(BA) \quad \text{and} \quad \operatorname{tr} A^t = \operatorname{tr} A.$$

**Theorem 4.21**

*To every operator  $\mathbf{A} \in \text{End}(\mathcal{V})$  are associated two intrinsic numbers,  $\det \mathbf{A}$  and  $\text{tr } \mathbf{A}$ , which are the determinant and the trace of the matrix representation of the operator in any basis of  $\mathcal{V}$ .*

## Corollary 4.22

*The determinant of a diagonal matrix  $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_N)$  is given by the product of its diagonal elements:*

$$\det D = \prod_{j=1}^N \lambda_j.$$

### Theorem 4.23

For any two matrices  $A, B \in \mathcal{M}^{N \times N}$ , the mapping

$$g : \mathcal{M}^{N \times N} \times \mathcal{M}^{N \times N} \rightarrow \mathbb{C}$$

defined by  $g(A, B) = \text{tr}(A^\dagger B)$  is a sesquilinear inner product.